Scenario 1: Account, Contact and Opportunity Lifecycle

Scenario:
Automate an end-to-end sales flow covering Account, Contact, and Opportunity creation and validation.

Expected automation:

1. **Login to Salesforce using Playwright.**
2. **Generate unique Account and Contact test data dynamically.**
3. **Create an Account using Salesforce API invoked through Playwright.**
4. **Store the generated Account ID and runtime values in a Map.**
5. **Create a Contact associated with the Account using Playwright API.**
6. **Create an Opportunity for the Account through Salesforce UI.**
7. **Update Opportunity Stage, Amount, and Close Date.**
8. **Store generated record IDs using Maps/Collections.**
9. **Execute SOQL through Playwright to retrieve Account, Contact, and Opportunity data.**
10. **Validate the Account → Contact → Opportunity relationships.**
11. **Validate key business data through both Salesforce UI and SOQL/API.**
12. **Clean up created test records through Playwright API.**
13. **On failure, capture the failing step, Salesforce record ID, expected/actual values, logs, and screenshot.**
14. **Run successfully for 5 consecutive executions.**

Scenario 2: Multiple Contacts and Opportunities with Dynamic Data

Scenario:
Automate an Account containing multiple Contacts and Opportunities and validate all dynamically generated relationships.

Expected automation:

1. **Login to Salesforce.**

2. **Generate unique Account data.**
3. **Create the Account using Playwright API.**
4. **Dynamically generate at least 3 Contacts.**
5. **Dynamically generate at least 3 Opportunities with different Stage, Amount, and Close Date values.**
6. **Store test datasets and generated Salesforce IDs using Maps/Collections.**
7. **Associate all records with the dynamically created Account.**
8. **Update selected Contact and Opportunity information through Salesforce UI.**
9. **Execute SOQL through Playwright to retrieve all related records.**
10. **Match returned records against the runtime Collection rather than relying on hard-coded values.**
11. **Validate updated values through both UI and SOQL/API.**
12. **Identify and report missing, duplicate, or incorrectly related records.**
13. **Clean up all dynamically created records.**
14. **Capture appropriate diagnostic information on failures.**
15. **Complete 5 consecutive successful executions.**

Scenario 3: Data-Driven Lead Creation and Conversion

Scenario:
Automate multiple Lead journeys using dynamic test data and condition-based Lead conversion.

Expected automation:

1. **Login to Salesforce.**
2. **Dynamically generate at least 5 Lead datasets with different combinations of fields such as Lead Source, Industry, Rating, and Country.**
3. **Maintain the datasets in a Collection.**

4. **Create Leads dynamically using Playwright/API.**
5. **Capture Lead IDs in a Map.**
6. **Based on predefined conditions in the test data, determine which Leads should be converted.**
7. **Convert eligible Leads through Salesforce UI.**
8. **Capture the resulting Account, Contact, and Opportunity IDs.**
9. **Maintain original Lead ID → Account → Contact → Opportunity relationships in Collections.**
10. **Execute SOQL through Playwright to validate converted and unconverted Lead states.**
11. **Validate generated Account, Contact, and Opportunity information.**
12. **Compare Salesforce UI results with backend values.**
13. **Allow processing/reporting of remaining datasets if one dataset fails.**
14. **Capture logs, screenshots, IDs, and expected/actual values for failures.**
15. **Clean up dynamically generated records.**
16. **Demonstrate 5 consecutive stable executions.**

Scenario 4: Multi-Level Account Hierarchy and Related Records

Scenario:
Automate a dynamically generated Salesforce Account hierarchy containing multiple Contacts and Opportunities.

Example:

Plain Text

Parent Account

|

+-- Child Account 1

| +-- Contacts

| +-- Opportunities

|

+-- Child Account 2

+-- Contacts

+-- Opportunities

Expected automation:

1. **Login to Salesforce.**
2. **Dynamically create a Parent Account using Playwright API.**
3. **Create multiple Child Accounts linked to the Parent.**
4. **Dynamically create multiple Contacts for each Child Account.**
5. **Create multiple Opportunities for the Child Accounts.**
6. **Maintain the complete hierarchy using nested Maps/Collections.**
7. **Navigate through the Account hierarchy using Salesforce UI.**
8. **Modify selected child records through UI.**
9. **Execute relationship-based SOQL through Playwright.**
10. **Validate Parent → Child Account relationships.**
11. **Validate Child Account → Contact/Opportunity relationships.**
12. **Compare Salesforce data against the nested runtime Collection.**
13. **Detect incorrect, missing, or unexpectedly associated records.**
14. **Clean up dynamically generated records in the correct dependency order.**
15. **Provide appropriate diagnostic information for failures.**
16. **Demonstrate 5 consecutive stable executions.**

Scenario 5: Complete Lead-to-Sales Lifecycle with Dynamic Validation and Recovery

Scenario:
Develop a full Salesforce business flow combining Lead, Account, Contact, and multiple Opportunity transactions, including dynamic data management, backend validation, error reporting, and cleanup.

Expected automation:

1. **Login to Salesforce.**
2. **Generate a unique test-execution ID.**
3. **Dynamically generate multiple Lead datasets.**
4. **Create the Leads using Playwright.**
5. **Store test data and generated record IDs in Maps/Collections.**
6. **Determine which Leads should be converted based on runtime test-data conditions.**
7. **Convert selected Leads through Salesforce UI.**
8. **Dynamically capture generated Account, Contact, and Opportunity records.**
9. **Add additional Contacts to the converted Account using Playwright API.**
10. **Create additional Opportunities dynamically with varying Stage, Amount, and Close Date.**
11. **Update selected Opportunities through Salesforce UI.**
12. **Maintain expected Lead → Account → Contact → Opportunity relationships in nested Collections.**
13. **Execute multiple SOQL queries through Playwright.**
14. **Validate the complete record relationships and key business values through both UI and backend.**
15. **Track every record generated during execution in a cleanup Collection.**
16. **On failure, capture:**

- **Test execution ID**
- **Salesforce record ID**
- **Input dataset**
- **Failed step**

- **Expected value**
- **Actual value**
- **Screenshot**
- **Relevant API/SOQL error or response**

17. Perform automated cleanup in the appropriate dependency order.

18. Execute the complete flow successfully 5 consecutive times.